Landing case: conditional DDPM attribution of a ten-day IP-block shock

Ten-site linear conditional panel

Abstract

The product answers one query. Ten days ago the IP-block feature of cohort A spiked. If that day is put back on its reference path, what is today's purchase distribution for cohort B, at which lag does the discrepancy peak, and which block carries it? The shipped object is a conditional-distribution counterfactual [Chernozhukov et al., 2013], scored by random-Fourier MMD^2 [Gretton et al., 2012, Rahimi and Recht, 2007] under a conditional DDPM [Ho et al., 2020]. On a ten-site linear panel with a known kernel, the validation path peaks at day 10 on the weekly site, the intervened block is the IP feature, and the isolated copy of the same site stays on the floor. The ratio of that score to the no-intervention baseline is not a stable effect size: it moves from 7.7 to 31.6 as the kernel width changes.

1 The query

Cohorts are sites. On a shared calendar, site s has an IP-block series $B_{s,t} \in \mathbb{R}$ and a purchase index $Y_{s,t}$. The business query fixes a spike day t_0 and a horizon $h = 10$:

> If B_{s,t_0} had followed the reference path, what would the conditional law of Y_{s,t_0+10} be?

The product returns five objects for that query: a point estimate of the distributional discrepancy and a bootstrap interval; the path $\{\Delta_h\}$ over horizons; the scale weights of the conditioner; the fraction of counterfactual states inside the reference support; and the five-layer falsification report of Section 7.

2 What the product claims

The intervention replaces the lagged IP block with the reference draw and keeps covariates at their observed values. The target is the conditional law

$$P_\theta\big(Y_{s,t} \mid B^{\text{cf}}_{s,t-1:t-K}, X_{s,t}\big),$$

the ceteris paribus conditional counterfactual of Chernozhukov et al. [2013]. A structural effect along a named channel is a different target [Pearl, 2009]. Risk control, recommendation, and social transmission are represented by one site-specific distributed lag. They are not separately identified, and the product does not label a channel. MMD has no physical unit. The number that may be shown next to a decision is the raw random-Fourier score together with its interval; the ratio to the no-intervention baseline is a calibration diagnostic. A conditioning vector outside the reference support does not enter the headline.

Table 1: Effective kernels. The lag-10 query has a unique primary site.

Site	Name	Kernel	Role at lag 10
0	weekly primary	bump at day 10, weight 0.95	primary driver
1	weekly secondary	same center, weight 0.50	secondary
2	daily driver	bump at day 3, weight 0.85	other query
3	immediate driver	bump at day 1, weight 0.80	other query
4	weekly weak	bump at day 10, weight 0.28	weak weekly
5	decay, recent	exponential, weight 0.55 at lag 1	L1 temporal construction
6	social overlap	no lag kernel; overlap × lag 2	L1 overlap construction
7	isolated	potential copy of site 0, transmission off	L1 isolation
8	null	flat	negative control
9	null, noisy	flat, scale 1.55	negative control

3 Score

Let P_{obs} be the frozen sample of purchases on the query day, and let $P_\theta(\cdot \mid \text{cf})$ be the DDPM sample under the reference history. With $D = 100$ random Fourier features of the Gaussian kernel [Rahimi and Recht, 2007, Gretton et al., 2012],

$$\Delta^{\text{cf}} = \text{MMD}^2_{\text{RFF}}\big(P_{\text{obs}}(Y_{s,t_0+h}),\, P_\theta(Y_{s,t_0+h} \mid \text{cf})\big). \tag{1}$$

The no-intervention baseline Δ^{base} uses the same sampler on the observed history. The normalized diagnostic is $\widetilde{\Delta}^{\text{cf}} = \Delta^{\text{cf}} / \max(\Delta^{\text{base}}, 10^{-3})$. The interval is the percentile interval of (1) under 400 resamples of the two finite samples, with the Fourier features held fixed. Bandwidth is the median pairwise distance, unless a sensitivity setting scales it.

4 Landing panel

The landing test is a ten-site linear conditional distribution. Forty-eight windows, forty days each, spike day $t_0 = 18$, additive spike of size dose × 1. The last twelve windows are the validation slice. The conditional mean is linear in a fourteen-day lag, a social term at lag 2, and three covariates:

$$\begin{aligned}\mu_{s,t} = \alpha_s &+ \sum_{k=1}^{14} \omega_s(k)\, B_{s,t-k} + \beta^{\text{soc}}_s \cdot \text{overlap}_s \cdot B_{s,t-2} \\ &+ \gamma^\top X_{s,t} + 0.15\, U_s.\end{aligned} \tag{2}$$

The conditional standard deviation is $0.48\,\sigma_s\big(1 + 0.15\,|\text{signal}_{s,t}|\big)$. U_s is unobserved. A bump kernel has its peak at a site-specific lag; a decay kernel puts its mass on recent lags. Isolation multiplies ω_s by zero after the kernel is recorded, so the IP spike is still seen.

At unit dose the population mean shift of site 0 at lag 10 is 0.95. Doses on the validation windows cycle through $\{0.40, 0.70, 1.00, 1.30\}$, so the average oracle shift on those windows is 0.807. Site 6 is the only social transmitter, and its overlap grid is $\{0.20, 0.45, 0.70, 0.92\}$ at a fixed dose of 1. Site 0 also has a varying overlap, and $\beta^{\text{soc}}_0 = 0$, so overlap is a negative control there.

The same conditioner shape accepts a live retail panel. A store is a site, a promo or a competitor shock is B_t, and the daily sales distribution is Y_t [Makridakis et al., 2022]. The landing test keeps the kernel known so that a wrong peak can be seen. A live panel does not have that oracle; the five layers below are the acceptance test that remains.

5 Model that ships

The landing call is the package. Nothing else is required to reproduce the card.

```
import ddpm_cf_attribution
result = ddpm_cf_attribution.run()
```

`result["primary_driver"]` is the site, the lag, and the block. `result["headline"]` is Δ^{cf}, the interval, and the path shape. The call writes the five-layer report next to the package.

The diffused variable is the scalar purchase index. A length-one series has no spatial U-Net, so the denoiser is a two-block MLP with adaptive layer normalisation, the scalar analogue of the adaptive group normalisation used by Dhariwal and Nichol [2021]. History reaches the denoiser only after a three-way softmax gate over lags 1–2 (hourly), 3–5 (daily), and 6–14 (weekly). The training loss is the noise-prediction loss of Ho et al. [2020], with the lag block dropped on ten percent of steps. The noise schedule is the cosine schedule of Nichol and Dhariwal [2021], sixty steps on the shipped retry. Site identity enters as an embedding. The isolation flag does not. The acceptance weight for a scale is the normalised change in the predictive mean when that lag bucket is reset to its training mean. The softmax is logged and is not the acceptance statistic, because a gate and a downstream weight are not separately identified. On site 0 the softmax was $(0.066, 0.045, 0.889)$ and the ablation weight was $(0.17, 0.15, 0.68)$. Both put the mass on the weekly bucket.

Before diffusion, a quantile-regression MVP [Koenker, 2005] must clear a dose-response concordance of 0.70. On the validation windows the gradient-boosting quantiles scored 0.85, and the run entered Stage 2. The first diffusion fit failed the Wasserstein clause of Section 7. One pre-registered retry, one hundred and twenty epochs at width 192, is the fit reported below. Its best validation noise loss was 0.397.

The counterfactual path replaces B_{s,t_0} with the pre-spike latent and leaves later innovations as observed. In this panel the block has no autoregressive feedback, so that replacement is the rollout. Each future day is then scored by (1).

6 What the validation windows showed

Headline window. One validation window with dose 1, site 0, horizon 10:

$$\Delta^{\mathrm{cf}} = 0.631, \qquad 95\%\ \mathrm{CI}\ [0.426,\ 0.852], \qquad \Delta^{\mathrm{base}} = 0.021, \qquad \widetilde{\Delta}^{\mathrm{cf}} = 30.3. \tag{3}$$

The counterfactual mean sits 1.009 below the observed mean. The reference-history support rate on the whole validation grid is 1.00 at a coordinate z-score of 4. A stress path that pushes the lag-10 distractor b_2 out by 4.5 standard deviations is localised by a leave-one-coordinate nearest-neighbour score on that coordinate; clamping it cuts the neighbour distance from 4.05 to 2.63. The real attribution path did not need that clamp.

Who moves today's purchases. Table 2 averages the twelve validation windows. The primary site is site 0. Its path rises and then falls, with the peak on day 10. Sites whose kernels sit on other lags peak on those lags and are flat at lag 10. The isolated copy of the weekly primary, site 7, has mean $\Delta^{\mathrm{cf}} = 0.028$ against 0.412 for site 0.

The top three sites by raw Δ^{cf} are 0, 4, and 1, the same three sites as the oracle. Sites 4 and 1 swap order (0.165 against 0.143; oracle 0.238 against 0.425). Normalising by Δ^{base} lifts null sites whose baseline is nearly zero. The driver call uses the raw score and the peak day together. Across windows, the signed mean gap on site 0 averages 0.794, against the oracle average 0.807.

Table 2: Validation path under the reference intervention. Oracle is the mean shift at lag 10, averaged over the dose cycle. Δ^{cf} is raw random-Fourier MMD^2.

Site	Oracle	Δ^{cf}	Peak day	Shape
0 weekly primary	0.807	0.412	10	peak, then decline
1 weekly secondary	0.425	0.143	10	peak, then decline
4 weekly weak	0.238	0.165	11	peak, then decline
5 decay	0.008	0.046	1	irregular
3 immediate	0.000	0.034	1	irregular
6 social overlap	0.000	0.028	2	peak, then decline
7 isolated	0.000	0.028	15	no lag-10 mass
8 null	0.000	0.026	4	on the floor
9 null, noisy	0.000	0.024	7	on the floor
2 daily driver	0.000	0.020	3	peak, then decline

Table 3: Frozen acceptance versus the validation run.

Layer	Rule	Measured	Call
L1	≥ 3 of 4 constructions	temporal -0.76; overlap concordance 0.87; isolation $d = -1.48$, $p < 0.001$; dose Spearman 0.67	pass
L2	placebo, peak in $\{9, 10, 11\}$, scale match	placebos $< 0.35 \times \Delta_{10}$; peak at 10; weekly, daily, hourly sites match	pass
L3	PIT $p > 0.05$, $W < 0.1\,\mathrm{sd}$, HSIC in $[0.8, 1.2]$	$p = 0.445$; $W/\mathrm{sd} = 0.245$; HSIC 1.009	fail
L4	≥ 3 proxies agree	CATE $+1.005$, mask gap $+0.369$, business peak on day 10, weekly gate	pass
L5	three CVs and E-value > 2	baseline CV 0.13; architecture CV 0.16; E-value 4.06; kernel CV 0.70	fail

Which block. Replacing the IP lag gives $\Delta = 0.412$. Resetting the three lag-10 distractors to their training means gives 0.043. Resetting the covariates gives 0.112. The block that carries the query is the IP feature.

Where the past information sits. Placebo horizons 3, 5, 15, and 20 score 0.058, 0.071, 0.049, and 0.072, against 0.412 at the true lag and 0.078 for the no-intervention baseline. The precision scan over days 1 through 15 peaks at day 10. Daily site 2 peaks at day 3, immediate site 3 and the decay site peak at day 1, and the overlap site peaks at day 2, which is the only lag in its structural term.

7 Five-layer readout

Thresholds were fixed before the run. A layer fails closed.

Relative rank. Dose concordance on site 0 is 0.80 and the Spearman correlation with dose is 0.67, short of the 0.70 clause, so that construction fails on its own. The decay site ranks horizons with Spearman -0.76. The same statistic on the weekly site is $+0.74$: a monotone-lag story is the

wrong assumption for a day-10 kernel, and that failure is charged to the construction. Overlap concordance on the social site is 0.87; on the risk site the overlap Spearman is -0.13. Isolation of the matched weekly site has paired Cohen's $d = -1.48$. Three constructions pass.

Observational calibration. The probability integral transform [Dawid, 1984] has Kolmogorov–Smirnov $p = 0.445$. The Hilbert–Schmidt dependence ratio [Gretton et al., 2005] between generated outcomes and the conditioner, over the same ratio for observed outcomes, is 1.009. Mean per-cell Wasserstein distance is 0.245 standard deviations of Y, above the clause of 0.10. A Gaussian oracle that sees only the features, putting the realised U into the variance, sits at 0.180 on the same comparison. Two samples from one conditional sit at 0.080. The 0.10 clause lies under the feature-only floor of this panel. The DDPM is 0.065 above that floor. PIT and HSIC pass. The layer still fails closed.

Proxies. An R-learner [Nie and Wager, 2021] with the lag-10 IP feature as the continuous treatment estimates a site-0 coefficient of $+1.005$. The rollout's mean gap has the same sign. The mask gap $\Delta^{\mathrm{cf}} - \Delta_{\mathrm{mask}} = 0.369$. The excess of the purchase index over the spike day peaks on day 10, the same day as Δ_h. Four proxies agree.

Fragility. Four reference replacements of the spike day, residual, mean, resample, and zero, give raw scores 0.412, 0.413, 0.533, and 0.416 (coefficient of variation 0.13). The DDPM, the site-interacted linear Gaussian, and the quantile MVP score 0.412, 0.408, and 0.308 (coefficient of variation 0.16). The E-value [VanderWeele and Ding, 2017] at Cohen's $d = 0.92$ is 4.06. Bandwidth multipliers $1/2$, 1, and 2 move the normalized diagnostic through 7.7, 13.9, and 31.6 (coefficient of variation 0.70). That clause fails. The stable objects are the ranking and the peak day.

8 Label on the decision

Ship the identity of the driver, not a unitless multiple. In this landing the driver is site 0, lag 10, IP block B, weekly scale. The purchase distribution moves with that cell: the dose-1 window scores 0.631 with interval $[0.426, 0.852]$, and the mean gap is about one purchase-index unit. The path is a rise into day 10 and a decline after it. Sites with the same spike and a closed kernel, including the isolated twin, do not move.

Put three labels on the card. The normalized ratio is a calibration check; at the default bandwidth it is about 30, and across the three protocol bandwidths it runs from 7.7 to 31.6. The Wasserstein clause of the observational layer fails against a threshold that sits below the unobserved-confounder floor; PIT and the dependence ratio pass, and the conditional mean gap (0.794 against an oracle of 0.807) is the calibration fact the card should quote. The E-value is 4.06. A randomised confirmation is still the next measurement if the decision isolates a live IP range. The product does not name a risk, recommendation, or social channel.

References

Victor Chernozhukov, Iván Fernández-Val, and Blaise Melly. Inference on counterfactual distributions. *Econometrica*, 81(6):2205–2268, 2013. doi: 10.3982/ECTA10582.

A. Philip Dawid. Present position and potential developments: Some personal views: Statistical theory: The prequential approach. *Journal of the Royal Statistical Society: Series A*, 147(2): 278–290, 1984.

Prafulla Dhariwal and Alexander Nichol. Diffusion models beat GANs on image synthesis. In *Advances in Neural Information Processing Systems*, volume 34, pages 8780–8794, 2021.

Arthur Gretton, Olivier Bousquet, Alexander Smola, and Bernhard Schölkopf. Measuring statistical dependence with Hilbert–Schmidt norms. In *Algorithmic Learning Theory*, volume 3734 of *Lecture Notes in Computer Science*, pages 63–77. Springer, 2005.

Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *Journal of Machine Learning Research*, 13:723–773, 2012.

Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, volume 33, pages 6840–6851, 2020.

Roger Koenker. *Quantile Regression*. Cambridge University Press, 2005.

Spyros Makridakis, Evangelos Spiliotis, and Vassilios Assimakopoulos. M5 accuracy competition: Results, findings, and conclusions. *International Journal of Forecasting*, 38(4):1346–1364, 2022.

Alexander Quinn Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 8162–8171. PMLR, 2021.

Xinkun Nie and Stefan Wager. Quasi-oracle estimation of heterogeneous treatment effects. *Biometrika*, 108(2):299–319, 2021. doi: 10.1093/biomet/asaa076.

Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2 edition, 2009.

Ali Rahimi and Benjamin Recht. Random features for large-scale kernel machines. In *Advances in Neural Information Processing Systems*, volume 20, pages 1177–1184, 2007.

Tyler J. VanderWeele and Peng Ding. Sensitivity analysis in observational research: Introducing the E-value. *Annals of Internal Medicine*, 167(4):268–274, 2017. doi: 10.7326/M16-2607.